Scientific notation

1

a 2.49 b $20\,000\,000$ c $41\,300$ d 0.009

e $897\,000$ f $50\,300$ g 3014 h $8\,000\,000$

i $13\,008\,000$ j 0.836 k $7\,910\,000$ l 0.00714

2

a 3 b 5 c -5 d 10^{-7}

3

a 3×10^{2} b 4.5×10^{4} c 7×10^{-2} d 6.23×10^{-3}

e 2×10^{3} f 2×10^{-3} g 1×10^{-4} h 8.84×10^{5}

i 8.4626×10^{7} j 6.14×10^{0} k 3.47×10^{-4} l 3×10^{-5}

m 7×10^{-2} n 7.64×10^{-1} o 2×10^{6} p 1×10^{-4}

4 **D**

5

a

i Yes ii No iii No iv Yes

v No vi Yes

b 4.5×10^{-4}

6

a

i 7.8×10^{2} ii 10

b

i 4×10^{-3} ii $10\,000$

c

i 9×10^{9} ii $30\,000$

7

a

i 7.27×10^{-7} ii 2.22×10^{-7}

b

i 1.25×10^{2} ii 1.25×10^{-6}

c

i 9.37×10^{-4} ii 6×10^{-9}

d

i 8.31×10^{-9} ii 5.13×10^{-9}

e

i 7.88×10^{2} ii 2.93×10^{-2}

f

i 8.99×10^{-3} ii 1.99×10^{-6}

g

i $4.7 \times 10^{9} \times 10^{8}$ ii $4.7 \times 10^{-8} \times 10^{8}$

h

i $2.12 \times 10^{7} \times 10^{2}$ ii $2.12 \times 10^{-7} \times 10^{4}$

8

a 6×10^{11} b 1.6×10^{-7} c 3×10^{3} d 3×10^{2}

e 4.5×10^{4} f 2.1×10^{10} g 2.4×10^{4} h 4.4×10^{5}

Scientific notation and calculators

9

a $581\,000$ b 0.000316 c $2\,500\,000$ d 0.00705

10

a 0.003844 b 0.013

11

a 1.120×10^{-15} b 3.342×10^{2}

12 At any particular time, approximately 2.32×10^{5} aircraft are registered as being in flight over 7.772×10^{10} square metres of air space.

Significant figures

13

a Three
b Four
c Four
d Three

14

a 5.0
b 0.0060
c $801\,600\,000$
d 0.0507

15

a
i 3
ii $422\ \mathrm{m}^3$

b
i 2
ii $2.5\ \mathrm{mL}$

16

a 2.14×10^9
b 1.98×10^{-5}

17 4549

18 512

19

a 3499
b 2500
c 999

20

a 1
b 1.09

Applications

21 The diameter of a human cell

22

a 1.332×10^9
b 1.332×10^{21} litres

23

a 2 million each second
b $120\,000\,000$

24

a 1 hundred thousandth of a metre
b $0.01\ \mathrm{mm}$

25

a 3×10^{-1} km/s b 3×10^{5} km/s c 1 000 000

26

a 2.5×10^{5} km^2 b 6.8×10^{7}

c 2.72×10^{2} per km^2

27

a 8×10^{5} metres per hour b 1000

28 3300

29 135 000 kg

30 2×10^{7} g

31 7.95×10^{7}

32 2000 kg

33

a 1595 g b 2×10^{26} atoms

34

a 2.107×10^{24} atoms b 9 g

c 1.495×10^{-23} g

35

a 29 998 800 m b 7×10^{8} m

36 1.4130×10^{9} days

37 1.25×10^{14}